Chapter 1
Introduction

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Keywords Underwater cultural heritage · Deep sea mining · Deep seabed mining · Deep-sea mining · Intangible cultural heritage · International seabed authority · Underwater archaeology

Underwater cultural heritage (UCH) is integral to both cultural heritage and natural heritage, thus making it a shared **Ocean Heritage**. The stories of our societies and our ancestors are wrapped in intangible connections with the ocean and preserved on the seafloor as artefacts, shipwrecks, and remains of those lost or buried at sea. Deep sea mining (DSM) threatens this Ocean Heritage and has the potential to destroy crucial pieces of our past and disrupt our deep cultural connections to the ocean. This book is designed to highlight the potential threat of DSM, covered here mostly in international waters and as regulated by the International Seabed Authority (ISA), and make ocean professionals across many different fields aware of those dangers and of the work being done to help mitigate the possible launch of DSM operations. It does not aim to be the final word on the issue, but rather start a discussion and encourage scientists and cultural leaders to engage and work together.

When beginning a project that sheds light on threats to our Ocean Heritage, we may simply look at those threats that currently exist, as has been done already in *Threats to Our Ocean Heritage: Bottom Trawling* (Jarvis, 2024) and *Potentially Polluting Wrecks* (Brennan, 2024). However, there are also potential dangers to Ocean Heritage—DSM is one of these, but one that many UCH professionals might not yet have considered.

Deep sea mining is a 'potential commercial industry with a goal to mine mineral deposits from the ocean floor, in the hopes of extracting metals such as manganese,

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C. Jarvis (ed.), *Threats to Our Ocean Heritage: Deep Sea Mining*,
SpringerBriefs in Archaeology, https://doi.org/10.1007/978-3-031-98238-5_1

copper, cobalt, and nickel' (Dobush & Warner, 2024, p. 4). The deep sea, defined as the ocean below 200 metres, was previously considered uninhabited and largely unimportant (if considered at all), but that has changed over the last half century as deep-water exploration has increased. It is an enormous spread of ocean (covering 65% of the Earth's surface and more than 95% of the biosphere) with a biodiversity akin to tropical rainforests (Danovaro et al., 2010). The deep ocean provides many critical ecosystem services, including fish and shellfish for food; products from organisms that can be used for medicines; climate regulation; and historical, cultural, social, educational, and scientific value for people worldwide (Thurber et al., 2014; Le et al., 2017; Armstrong et al., 2012).

The deep sea is often referred to as earth's final frontier, but its inaccessibility means it receives relatively less attention than other ocean areas. The nineteenth century brought some of the first serious scientific interest in the ecosystem and what lived in it (Cuyvers et al., 2018). As early as the 1870s, researchers on HMS *Challenger* collected samples of polymetallic nodules (fist-sized mineral accretions) (Cuyvers et al., 2018). By the 1960s, these nodules were studied in more detail and the ocean was cited as a potential site to mine for the minerals inside. By the 1970s, companies were beginning to test mining methods, but the idea was shelved when metal demand dropped (Cuyvers et al., 2018).

But the deep sea has been important and vital to humans long before scientists entered the picture. Many cultural groups around the globe have creation stories linked to the ocean, and especially the deep ocean. The Haida, an indigenous group based for at least 12,500 years in the Haida Gwaii archipelago off the north Pacific coast of Canada, link the supernatural being SGaan Kinghlas, to the formation of a seamount within Canada's Exclusive Economic Zone (EEZ) (Fedje & Mathewes, 2005). In Africa, some cultures see the sea as their ancestral spirits' home and the ocean as a sacred place, with respect for it increasing with its depth. In South Africa, the Nguni-speaking diviner-healers cite travel underwater with spirits as a source of their power. They refer to the sea as '*Komkhulu*' (The Great Place) and some Zulu see it as a place where the water divinities live (Bernard, 2010). The deep sea has also had long-held significance in cultures all over as a habitat for marine species, such as whales, sharks, or turtles, which hold deep cultural importance (see Chap. 8, this vol. for more details).

DSM has the potential to cause major disturbances to the marine environment, including direct damage to the benthic fauna, habitat destruction, pollution from sediment plumes and wastewater discharge, and non-stop noise and light pollution across the water column (Weaver et al., 2018; Jones et al., 2018). Scientists expect that DSM would have devastatingly wide and irreversible impacts on microorganisms, fisheries, food chains, cultural traditions, and more by impacting the seafloor and water column (Jaeckel et al., 2023; Dobush & Warner, 2024) (Fig. 1.1).

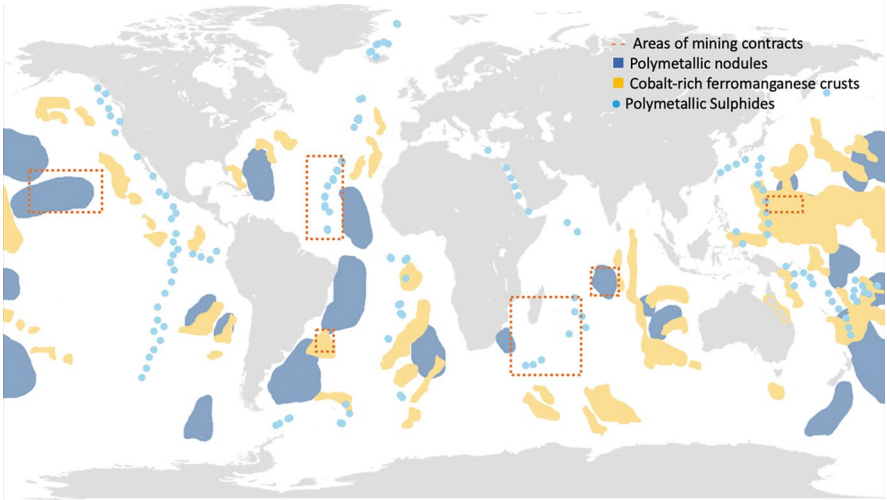

Fig. 1.1 Map of the elements for proposed mining as well as the zones in which the ISA has issued contracts. The Clarion Clipperton Zone, left side of the map, has been the most sought-after with 17 contracts issued for that area alone. (Source: Created by author, 2024. Based on information from the International Seabed Authority)

1.1 What Are Seabed Minerals?

Minerals are deposited and formed on and within the seabed in various ways, including as polymetallic nodules, polymetallic massive sulphides, and cobalt-rich ferromanganese crusts. The main areas of mineral deposits that commercial mining is focused on are located on the abyssal plains (for nodules), hydrothermal vents (for sulphides), and seamounts (for crusts).

Polymetallic nodules, sometimes called manganese nodules, occur primarily at depths below 4000 m and are found on the seafloor rather than embedded in hard crust (Fig. 1.2). They range in size from 4 to 10 cm in diameter and take millions of years to grow to the size targeted by mining (Cuyvers et al., 2018; Miller et al., 2018). Companies seeking to start the industry have a focus on minerals located in the Clarion Clipperton Zone (CCZ) in the Pacific, in the deep abyssal plain between Hawaii and Central America, as this area has a high concentration of nodules. DSM proponents point to the high concentration of multiple sought-after minerals in each nodule (Muñoz-Royo et al., 2021). The nodules 'provide the rare commodity of hard substrate' on the seafloor and are used as important habitat for immobile animals such as sponges, corals, or anemones on the abyssal plain to grow on (Stratmann et al., 2021, p. 1).

Polymetallic massive sulphides, or seafloor massive sulphides, are formed at hydrothermal vents where seawater meets magma (Fig. 1.3). The vents are biomass hot-spots as species use the chemical-rich liquid escaping from the vents as a source of energy rather than sunlight (chemosynthesis) (Van Dover et al., 2018). These vents 'provide habitat for novel faunal assemblages that have altered our views of

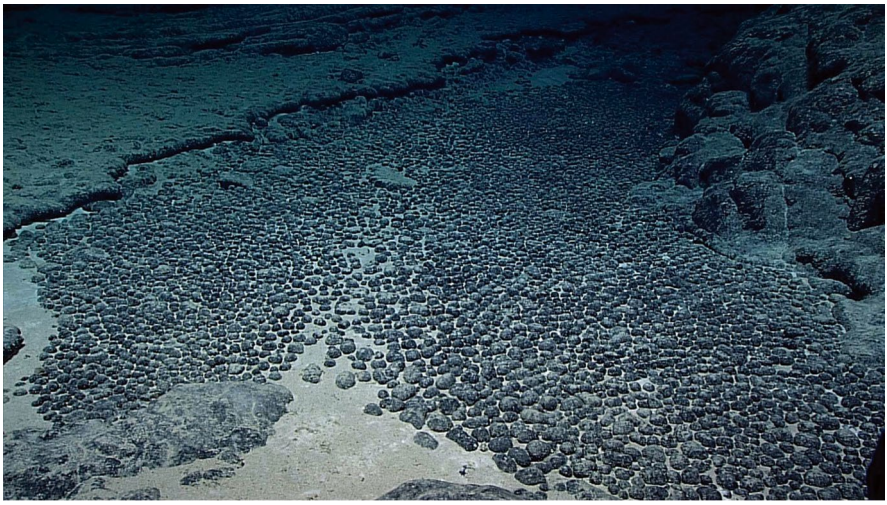

Fig. 1.2 A field of manganese nodules. (Source: NOAA Office of Ocean Exploration and Research)

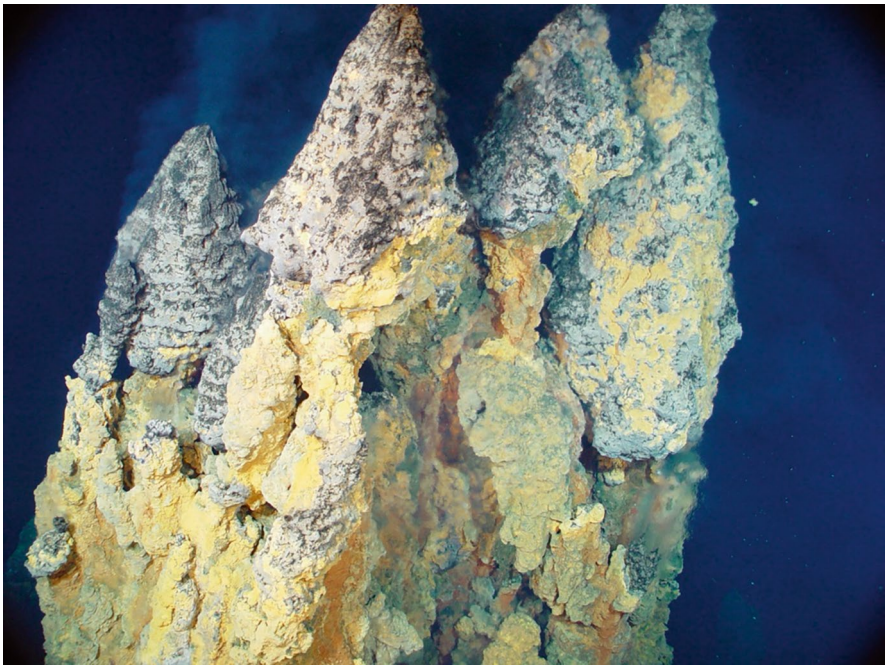

Fig. 1.3 Active smoker chimneys precipitating iron, copper, and zinc sulphides. (Source: NOAA Office of Ocean Exploration, Pacific Ring of Fire 2004 Expedition)

the primary energy sources and origins of life and exhibit substantial local and regional variation in structure and connectivity' (Smith et al., 2020, p. 853). Active hydrothermal vents 'host high-temperature (350 °C) "black smokers" that discharge metal-rich fluids' (van Dover et al., 2018, p. 21). The metals that precipitate form the polymetallic (especially copper and zinc) sulphides which are the inactive vent sites targeted by the DSM industry. The 'general perception of inactive sulphide deposits as biologically depauperate' means there has not been much exploration or biological characterisation of them, though there are some studies that show corals, sponges, or other species at inactive sites (Levin et al., 2016, p. 251; Van Dover et al., 1990; Erickson et al., 2009; Boschen et al., 2016).

Cobalt-rich crusts, or ferromanganese crusts, form on the slopes of seamounts and other underwater formations (Miller et al., 2018) (Fig. 1.4). The definition of seamounts can vary, but in the deep sea, they generally are 'isolated topographic elevations with summit depths at least 100 m above the seafloor' and are often volcanic in origin (Rogers, 2018, p. 137; Yesson et al., 2011). They support diverse ecological communities which 'take centuries or more to grow and are home to long-lived animals with complex interdependent relationships' (DOSI, 2023b, p. 2).

The metals targeted by DSM proposals are largely posited by mining companies as providing the building blocks necessary to decarbonise or move away from fossil fuels through batteries for things like electric vehicles (EVs) or as 'critical minerals' for military or other potential uses. The depletion of terrestrial deposits has also been cited as a reason to push forward with DSM (Filho et al., 2021). However, some new and next generation batteries use metals that are not targeted by DSM. As electric vehicle battery technology advances, there is a shift towards replacing batteries dependent on cobalt, nickel, and manganese with ones that require fewer minerals that need to be mined from the sea, such as lithium iron phosphate or sodium-ion batteries (Everett et al., 2023). The growth of lithium iron phosphate (LFP) batteries, which do not require minerals targeted by DSM, has been rapid, capturing a

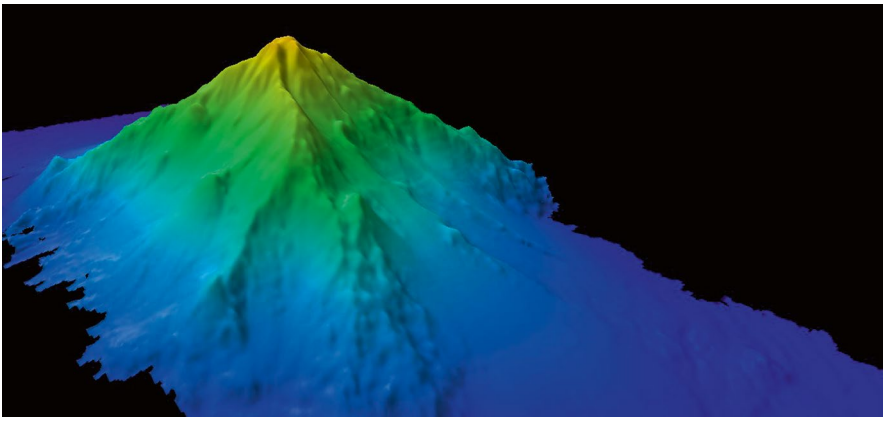

Fig. 1.4 A ~4200-m high seamount mapped during a NOAA expedition. (Source: NOAA Office of Ocean Exploration and Research, Mountains in the Deep: Exploring the Central Pacific Basin)

significant portion of the global EV battery market. Additionally, investment in the circular economy is directing focus to recycling and reuse of existing materials.

At the time of writing, commercial DSM in international waters has not begun, and there are still many unknowns as to how commercial scale mining could proceed (see Appendix I for ISA-issued exploration contracts and zones). Mining polymetallic nodules would require a machine, some companies' technology resembles a three-story-tall tractor, which is taken to the seafloor to take up the top layer of sediment containing the nodules with the targeted minerals; this process would also recover any other items on the seafloor such as rocks, flora, and small fauna (Fig. 1.5). These materials are transported up to a surface vessel where the slurry is processed and unneeded material is returned to the water via a discharge plume. The impacts of this method have been likened to strip mining. Another company has proposed a technology which would use artificial intelligence (AI) to pluck each nodule individually from the seafloor and posit this method as being less damaging to the seafloor habitats (Impossible Metals, 2024), despite the nodules themselves being critical habitats.

1.2 The Importance of Our Ocean Heritage

1.2.1 Underwater Cultural Heritage

Recognition of the potential devastation of UCH by DSM practices lags far behind the research conducted on natural heritage. The impacts of DSM could encompass both destruction of physical objects on the seafloor as well as damages to cultural practices and connections to the ocean. In the Pacific especially, DSM is related to the island environment because 'the ocean is at the heart of one's identity' and 'integral to the core identity of Islanders' (Tilot et al., 2021, p. 4). Other important elements of UCH in the Pacific include canoe and wayfinding practices, which are integral to cultural identity in Oceania. Tangible evidence of this is rare, but includes spiritual associations with natural heritage, such as shark calling (Jeffery et al., 2021). Additionally, vessels and shipwrecks represent important artefacts of human history and are the final resting place for many that have sunk in the open water areas of the high seas, including those from World Wars (Chap. 6, this vol.) and the Atlantic Slave Trade (Turner et al., 2020).

1.2.2 Natural Heritage[1]

Deep sea mining presents many risks to the cultural environment, but it is also anticipated to cause impacts at all levels of ocean ecosystems. The nature of impacts foreseen from DSM are numerous and can include direct or more indirect impacts.

[1] Thank you to Maila Guilhon for her input in this section.

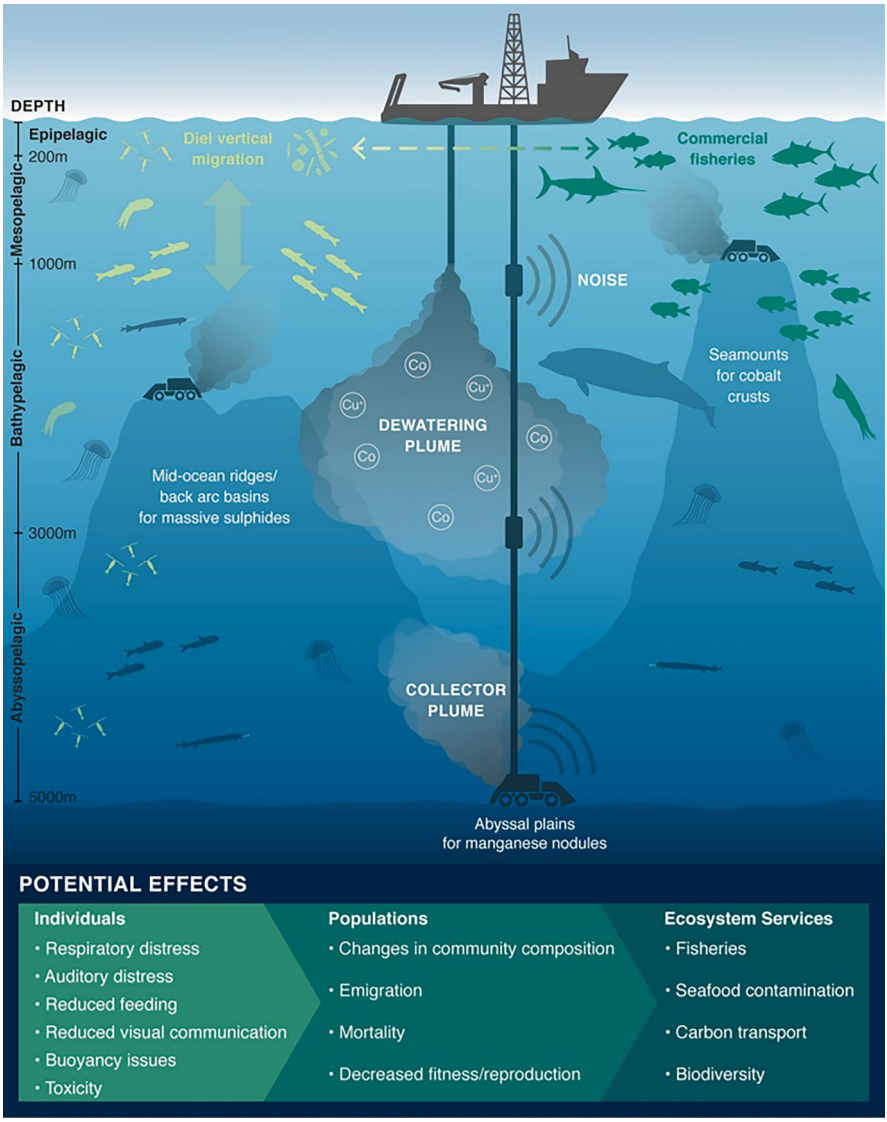

Fig. 1.5 Potential impacts of mining-generated sediment plumes (organisms and plume not to scale). (Source: © Drazen et al., 2020, PNAS. Graphic Artist: Amanda Merritt. Reprinted with permission from artist)

Direct impacts would be those that result from the extraction of the mineral structures and the destruction of biodiversity that live on or around it, the destruction or modification of habitats (Jenner et al., 2023). Indirect impacts are more diffuse, uncertain, and difficult to predict. These include, for instance, noise and light pollution and the formation of sediment plumes arising from machinery activity. Other indirect impacts can result from the discharge of return-water through pipes in the

midwater column, affecting deep-sea and pelagic ecosystems (Jenner et al., 2023). The deep ocean is considered a pristine ecosystem, vital to global health and climate stability with rich but yet poorly understood ecosystems—including the structures, functions, and the quantification of benefits arising from it. While several gaps and uncertainties on scientific knowledge remain, growing pressure for concluding negotiations of an exploitation code at the ISA (Singh et al., 2021) have led to an increasing call for a moratorium, a precautionary pause of DSM to allow for scientific knowledge about the deep-sea and the implications of DSM, including its social and economic implications (Singh et al., 2025; see Chap. 4, this vol.). Despite all of the above, a key question remains, at what point will knowledge be considered enough or sufficient to inform decisions if and how to move forward with DSM?

DSM activities are foreseen to take place kilometres deep in the ocean, in ecosystems largely unknown to science. In general, deep-sea ecosystems and habitats are characterised by slow dynamics and processes. In practice, that means that everything happens at a very slow pace in the deep ocean, including growth, reproduction and recovery. Due to that, in the occasion of impact, for example, from DSM activities, full ecosystems may not completely recover. As an example, a perturbation experiment that took place more than 30 years ago in the Pacific Basin has shown very low recovery rates (Simon-Lledó et al., 2019). Potential impacts from DSM are not restricted to the ocean floor. In fact, the impact of organism in the deep midwater (from 200 m to 5 km deep) has raised concern to scientists, given that this environment represents 'more than 90% of the biosphere, contain[s] fish biomass 100 times greater than the global annual fish catch, connect[s] shallow and deep-sea ecosystems, and play[s] key roles in carbon cycling, nutrient regeneration, and provisioning of harvestable fish stocks' (Drazen et al., 2020, p. 17456).

Deep-sea ecosystems provide a number of services and societal benefits (Le et al., 2017; Thurber et al., 2014; Tilot et al., 2021). Deep-sea ecosystems, like seamounts, play a very important role in the maintenance of important fishing stocks (see Gianni, 2004; Ramierez-Llodra, 2020). Consequently, an increase in pressure over seamount ecosystems through mining, for instance, could lead to impact in other sectors of commercial activities. Currents distribute heat around the planet which drives weather systems. Nutrients, vital to marine food chains, are regenerated at the ocean's surface by a process of upwelling, that bring new nutrients from deeper waters up to the surface. Moreover, the deep-sea is the largest carbon sink on earth with microorganisms and sediments helping to sequester some 25% of all carbon dioxide emissions since the mid-1990s (Hilmi et al., 2023; Gruber et al., 2019; Watson et al., 2020), having an important role in regulating the planet's temperature. The deep ocean also houses many migrating species, including whales, which have cultural value for many indigenous communities (Tilot et al., 2021). Finally, minerals structures such as polymetallic nodules serve as habitats by providing hard-subtract for organisms to fix on.

As the minerals found on the seafloor derive from different process of formation and deposit, foreseen technics to extract such minerals are very different. Although such technologies are yet under development, manganese nodules are likely to be

dredged off the seafloor, whereas SMS and cobalt crust extraction will utilise rock-cutting technology. Based on such, the main expected impacts into deep-sea ecosystems and biodiversity are listed by the IUCN (2022) as follows:

Disturbance of the seafloor The most direct impact of DSM and digging and gauging of the ocean will result in fragmentation or loss of ecosystem structure and function.

Sediment plumes Fine sediments will be disturbed on the seafloor creating plumes of suspended particles exacerbated by mining ships discharging wastewater at the surface. Such plumes, potentially dispersed for hundreds of kilometres could smother animals, harm filter-feeding species, and block visual communication.

Pollution Species such as whales, tuna and sharks maybe affected by noise, vibrations and light pollution caused by mining equipment and surface vessels, as well as potential leaks and spills of fuel and toxic products.

Recovery dynamics Several studies highlight that the deep seabed may not naturally recover from the impacts of DSM activities (Amadi & Mosnier, 2023; De Jonge et al., 2020).

Historically, deep sea areas have been understudied (Costello et al., 2010) with only 0.00001% investigated (Danovaro et al., 2020). Although over the years there has been an increase in deep-sea research and knowledge, the deep ocean remains an understudied environment that we know very little about. From a single research expedition at a contract region at the CCZ, which holds 17 exploration contracts at the ISA, of the more than 5000 species sampled—88% of the species examined were previously undescribed (Rabone et al., 2023). As another example, a recent discovery has demonstrated that polymetallic nodules may have the capacity to produce oxygen through a process of electrolysis (Sweetman et al., 2024). These are good examples reflecting how limited our knowledge about the deep sea and its biodiversity and structure is, consequently, how we know very little about how it works. Such extent of uncertainties requires the adoption of a precautionary approach into decision-making at the ISA.

Two recent studies provide an assessment of how available information on deep sea ecology matches up to the challenge of making wise judgements in the face of complexity and inherent uncertainty. The first addressed, in broad terms, the requirements of widely accepted standards for environmental impact assessment and associated management plans. The conclusion was that for the majority of matters that would be considered, there is neither adequate baseline data nor reliable information on impacts (Amon et al., 2022). A White Paper produced by the World Economic Forum (Yeh & WEF, 2022) referenced this work and extended the question to whether, given the current state of knowledge, DSM minerals can be included in responsibly sourced supply chains. The conclusion again pointed to very large knowledge gaps and also weakness in stakeholder engagement. It was found that the

potential effects that are currently the least predictable are the ones that most directly affect people and planet. Both studies touched on what effort might be required to close these gaps. Expert estimates ranged greatly but few suggested less than ten years of highly concentrated work, and many predicted much longer timescales.

1.2.3 Underwater Cultural Heritage as Both Natural and Cultural

The Pacific Ocean, in particular, serves as an important case study in which the natural and cultural heritage of the ocean is linked and of 'profound cultural, social, spiritual, and economic significance to Pacific Island communities' (Tilot et al., 2021, p. 2). This region has a high number of potentially polluting wrecks (PPWs), shipwrecks 'containing a cargo or a large volume of its own fuel that remains within the wreck and has the potential to pollute due to petroleum cargoes or bunkers' (Brennan, 2024, p. 1; further information see Chap. 6, this vol.). WWII especially left a large footprint in the Asia-Pacific region comprising thousands of shipwrecks, plane wrecks, and other items from both sides (Jeffery et al., 2021). Many of these World War wrecks sit in areas that are already a focus of seabed mining, as seen in Chap. 6. The risk of hitting one of these already fragile vessels and causing a massive oil leak is high and must be considered.

1.3 Societal Impacts of Deep Sea Mining

There are also potential risks for threats to human lives and worker safety associated with deep sea mining as well as socio-economic problems and impacts to coastal livelihoods. Large-scale development of DSM may cause a loss of access to coastal sites of cultural significance or those that are vital for fishing, tourism, or other economic activities (UNEP FI, 2022). Additionally, there is a risk of impact to human health through disruption to marine biodiversity which may impact food safety, security, and an acceleration of global climate change (Hamley, 2022).

Many studies have pointed to alternative sources for the minerals targeted in seabed mining. Miller et al. (2018) summarise the options as 'substituting metals in short supply, such as rare earths, for more abundant minerals with similar properties; landfill mining; and collection and recycling of components from products at the end of their life cycle' (p. 19). They also suggest promoting responsible e-waste recycling and a shift to a focus on reducing consumption. The European Academies Science Advisory Council (EASAC) found that 'the pressure for mining is driven by industry and economic interests rather than demands from the transition to a green economy' (EASAC, 2023, p. 17).

There are countless uncertainties regarding DSM. Although many companies have proposed technology to mine the minerals, at the time of writing, none have

publicly published outlined measures for the processing, storage, transportation, port transfers, or land infrastructure to conduct the operation. But any transport will likely have the same negative impacts as seen in other forms of maritime transport, including 'marine pollution, atmospheric emission, underwater noise and physical disturbance' (UNEP FI, 2022, p. 26). Additionally, equality of revenue and benefit sharing has not been guaranteed. Some ISA discussions have focused on financial sharing, but no consensus has been reached (Pickens et al., 2024). 'Rights of Nature', a theory that cites the inherent rights as associated with the ecosystem and species similarly to the concept of human rights, have not been well considered either (Miller et al., 2021). Miller and colleagues argue that while 'the concept of giving rights to nature might seem new to many in the industrialised West, the concept is not unusual to some Indigenous communities' (Miller et al. 2021, p. 5). Ecuador, for example, includes Rights of Nature in its constitution and this could be applied to the ocean alongside a precautionary principle (see Chap. 2, this vol.).

At the time of writing, there are voices speaking out against DSM in all sectors. Businesses and financial institutions including Apple, Microsoft, Google, Samsung, SAP, Phillips, ABN AMRO, BBVA, Cooperative Banking, Lloyds Banking Group, NatWest, Credit Suisse, Standard Charter Bank, Triodos Bank, and Patagonia have voiced support for a DSM moratorium or pledged to cut deep sea metals out of their supply chains.[2] Some in the automotive industry have also come out against DSM including BMW, Volvo, Volkswagen, Daimler, General Motors, Renault, and Ford.[3]

[2] For an overall discussion on companies against DSM see: WWF. (2021, December). *Business Backlash Against Deep-Sea Mining Grows*. https://wwf.panda.org/wwf_news/?4541466/Business-Backlash-Against-Deep-Sea-Mining-Grows; Greenpeace. (n.d.). *Race to the Top*. Stop Deep Sea Mining. Retrieved January 22, 2024, from https://www.racetothetop.app; Deep Sea Conservation Coalition. (n.d.). Voices calling for a moratorium: Companies and finance. Retrieved July 8, 2024, from https://deep-sea-conservation.org/solutions/no-deep-sea-mining/momentum-for-a-moratorium/companies-and-finance/; and Reid, H. (2021, March 31). Google, BMW, AB Volvo, Samsung back environmental call for pause on deep-sea mining. *Reuters*. https://www.reuters.com/business/sustainable-business/google-bmw-volvo-samsung-sdi-sign-up-wwf-call-temporary-ban-deep-sea-mining-2021-03-31/. For company specifics see: Microsoft. (2021). *Building a supply chain with integrity, accountability and respect* (Responsible Sourcing Report FY21). Microsoft Devices; ABN AMRO. (2021). *Exclusion List*; BBVA. (2020). *Environmental and Social Framework*; The Cooperative Bank. (2022). *Our customer-led Ethical Policy*; Hicks, R. (April 2022). Credit Suisse joins growing list of banks shunning deep-sea mining. EcoBusiness. https://www.eco-business.com/news/credit-suisse-joins-growing-list-of-banks-shunning-deep-sea-mining/; Standard Chartered. (2023). *Position Statement—Extractive industries*; Tridos Investment Management. (December 2021). *Tridos Bank supports moratorium on deep-sea mining*; and Patagonia. (December 2021). *Patagonia Calls for Moratorium on Seabed Mining*.

[3] For an overall discussion on companies against DSM see: WWF. (2021, December). *Business Backlash Against Deep-Sea Mining Grows*. https://wwf.panda.org/wwf_news/?4541466/Business-Backlash-Against-Deep-Sea-Mining-Grows; Greenpeace. (n.d.). *Race to the Top*. Stop Deep Sea Mining. Retrieved January 22, 2024, from https://www.racetothetop.app; Deep Sea Conservation Coalition. (n.d.). Voices calling for a moratorium: Companies and finance. Retrieved July 8, 2024, from https://deep-sea-conservation.org/solutions/no-deep-sea-mining/momentum-for-a-moratorium/companies-and-finance/; and Reid, H. (2021, March 31). Google, BMW, AB Volvo, Samsung back environmental call for pause on deep-sea mining. *Reuters*. https://www.reuters.com/

Fig. 1.6 Growing list of nations that have opposed DSM as of July 2025. (Source: Deep Sea Conservation Coalition, 2025. Reprinted with permission)

Other voices against DSM include more than 900 marine science and policy experts from more than 44 countries,[4] the European Parliament,[5] the Pacific Parliamentarians Alliance on Deep-Sea Mining,[6] the Parliament of the Canary Islands,[7] and the Regional Parliament of the Azores.[8] These voices are all echoing the sentiments of the 37 countries that have said no to DSM (Fig. 1.6).

Underwater cultural heritage has largely been left out of discussions about DSM's environmental impact while intangible cultural heritage has often been disregarded. There is a lack of baseline data to use as a benchmark to monitor DSM operations, and rules have yet to be developed for keeping scientists or archaeologists involved with their operations if they begin. As the archaeology chapter (Chap. 5) of this volume shows, telepresence and transmission of data is vital to studying deep-water environments and Ocean Heritage.

business/sustainable-business/google-bmw-volvo-samsung-sdi-sign-up-wwf-call-temporary-ban-deep-sea-mining-2021-03-31/. Ford has agreed to refrain from using deep sea minerals in the supply chain but has not announced support for a global moratorium (Greenpeace, *Race*).

[4] Deep Sea Conservation Coalition. (n.d.). Voices calling for a moratorium: Scientists. Retrieved July 8, 2024, from https://deep-sea-conservation.org/solutions/no-deep-sea-mining/momentum-for-a-moratorium/scientists/.

[5] Deep Sea Conservation Coalition. (n.d.). Voices calling for a moratorium: Governments and parliamentarians. Retrieved July 8, 2024, from https://deep-sea-conservation.org/solutions/no-deep-sea-mining/momentum-for-a-moratorium/governments-and-parliamentarians/.

[6] Ibid.

[7] Ibid.

[8] Ibid.

1.4 Steps Forward

The deep sea remains an area of the ocean in need of further study and scientists face large gaps in their understandings of its ecosystems let alone the UCH there. This makes it impossible to properly assess the potential impacts of DSM operations and as they stand, current ISA regulations are not sufficiently protective of Ocean Heritage. Global networks are hesitant to endorse the practice with the UN Environment Programme Finance Initiative (UNEP FI) concluding that financing DSM is inconsistent with principles of sustainable finance due to its significant reputational, regulatory, and operational risks. The International Capital Market Association (ICMA) issued guidance on sustainable blue economy bonds, explicitly excluding DSM. Similarly, the UN Conference on Trade and Development (UNCTAD) has excluded DSM from its sustainable ocean economy classification due to its high environmental risk.

Humans have a long history of detrimental activities in the ocean, but the deep sea represents 'one area where it is not too late and where we have a chance to get it right' (Cuyvers et al., 2018, p. vii). This point holds especially true when archaeologists consider the impacts of bottom trawling on their sites and the notable destruction this unregulated practice has caused (see Jarvis, 2024). With DSM, there is an opportunity to intervene before operations start. This was not successfully done with trawling and countless pieces of history have been lost. These mistakes cannot be repeated with DSM.

Tilot et al. (2021) explore current legal frameworks in the Pacific and 'an analysis of some traditional dimensions and cultural/societal aspects of marine resource management' to find that:

> the policies and practices developed in the Pacific could well serve as a suitable model elsewhere to reconcile commercial, ecological, cultural and social values within the context of deep sea mineral exploitation in addition to sustaining the Human Well-being and Sustainable Livelihoods (HWSL) of the Pacific communities and the health of the Global Ocean (p. 1, 2).

The authors note that many traditional marine management practices have flexible restrictions and can be applied in conjunction with other science-based tools such as Environmental Impact Assessments or permitting. Traditional knowledge and community-based marine management should be considered useful tools with a key role in this discussion (Tilot et al., 2021). These conversations traditionally have not found a place in the regulatory development at the ISA (Jarvis et al., 2023).

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